

INTRODUCTION TO DATA WAREHOUSING

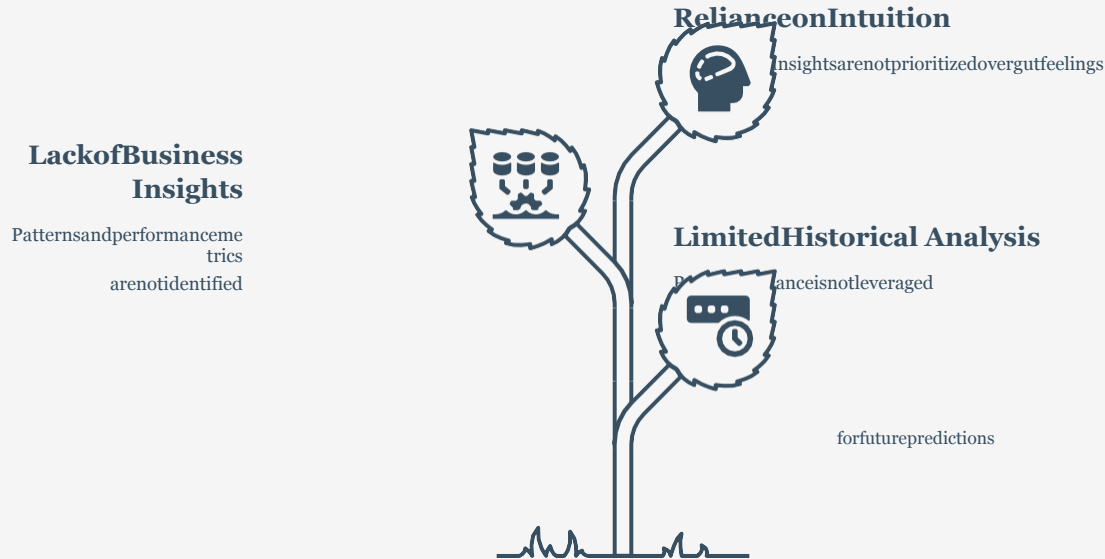
The Data Warehousing Ecosystem: From Raw Data
to Strategic Decisions



Data Warehousing Ecosystem

The ecosystem spans from raw data to strategic decisions.

DecisionSupportSystem(DSS)



Purpose: Enabling Strategic Decision Making through Historical Analysis

- Strategic Decision Making: Long-term planning and goal setting.
- Business Analysis: Identifying patterns and performance metrics.
- Historical Data: Leveraging past performance to predict future outcomes.
- Better Decision Making: Data-driven

riven insights over intuition.

Operational(OLTP)vs.DecisionSupport(OLAP)

Operational System(OLTP)

**Daily
transactions**

Focusesondaily
transactionsandcurrent

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data.

Fastprocessing

Optimizedforfast
processingandhigh-frequency
updates.

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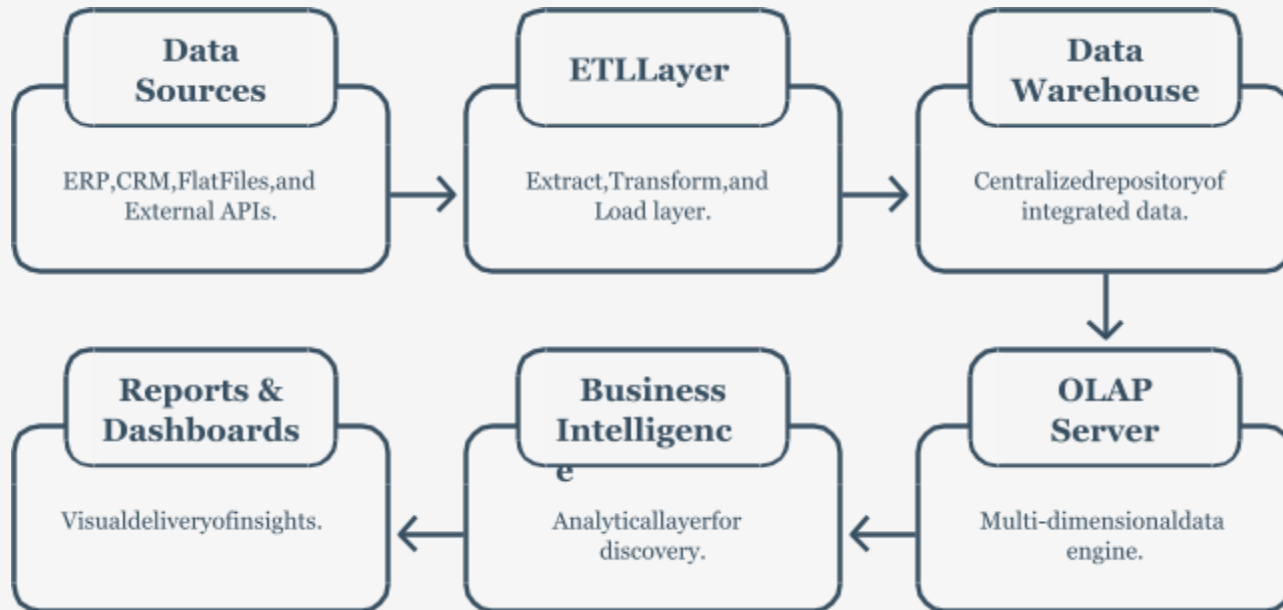
Historicaldata

Focusesonhistoricaldata
andbusinessintelligence.

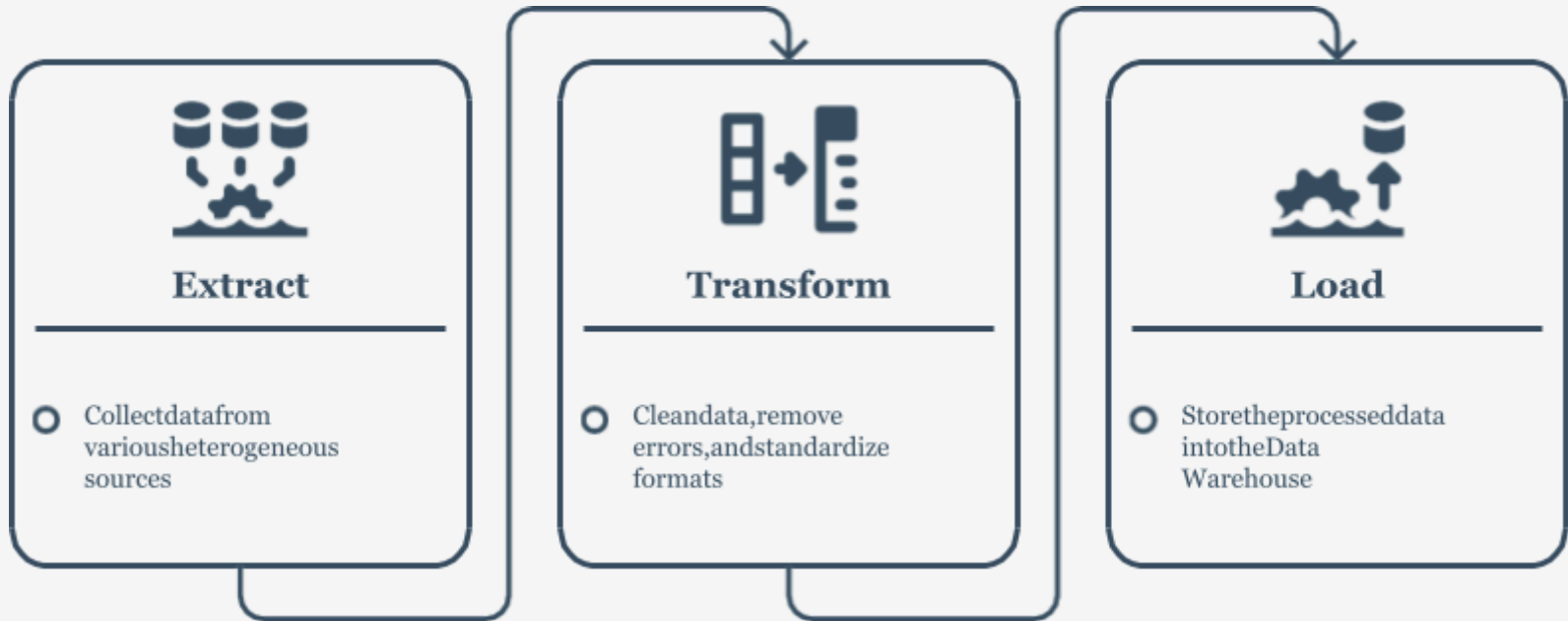
Dataanalysis

Optimizedfor
dataanalysis,
reporting,
and trend
analy-sis.

Data Warehouse Architecture: End-to-End Flow



The ETL Process



OLAP: Online Analytical Processing

Enabling Multi-dimensional Analysis and Fast Querying

& Decision Support: Providing the evidence for business choices.

& Business Intelligence: The core engine of modern BI.



Supports multi-dimensional analysis and fast querying



OLAP
OLTP



Focuses on transactional processing and real-time updates

The Data Cube



Dimensions

Dimensions: Time, Product, and Location.



Slice&Dice

Selecting specific sub-sets of data.



DrillDown&RollUp

Changing the level of detail/granularity.



Pivot

Re-orienting the view of the cube.

3-Dimensional Analysis and
Core Operations

Dimen-si

Fast Query
Performance

 Slower Query
Performance

onalData

Modeling

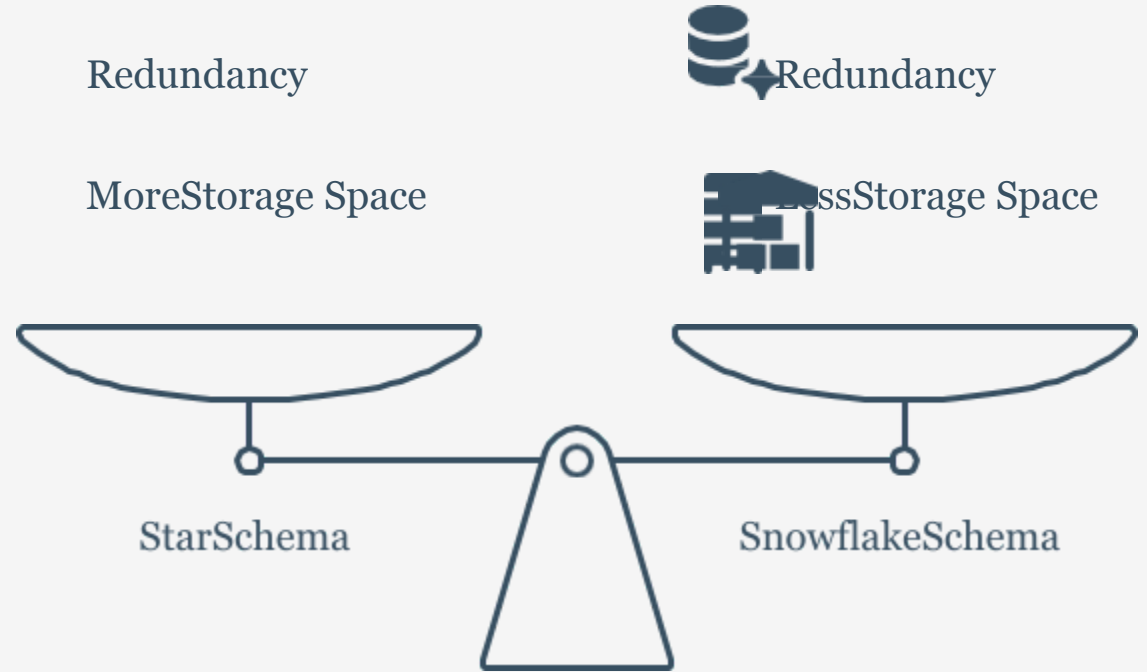


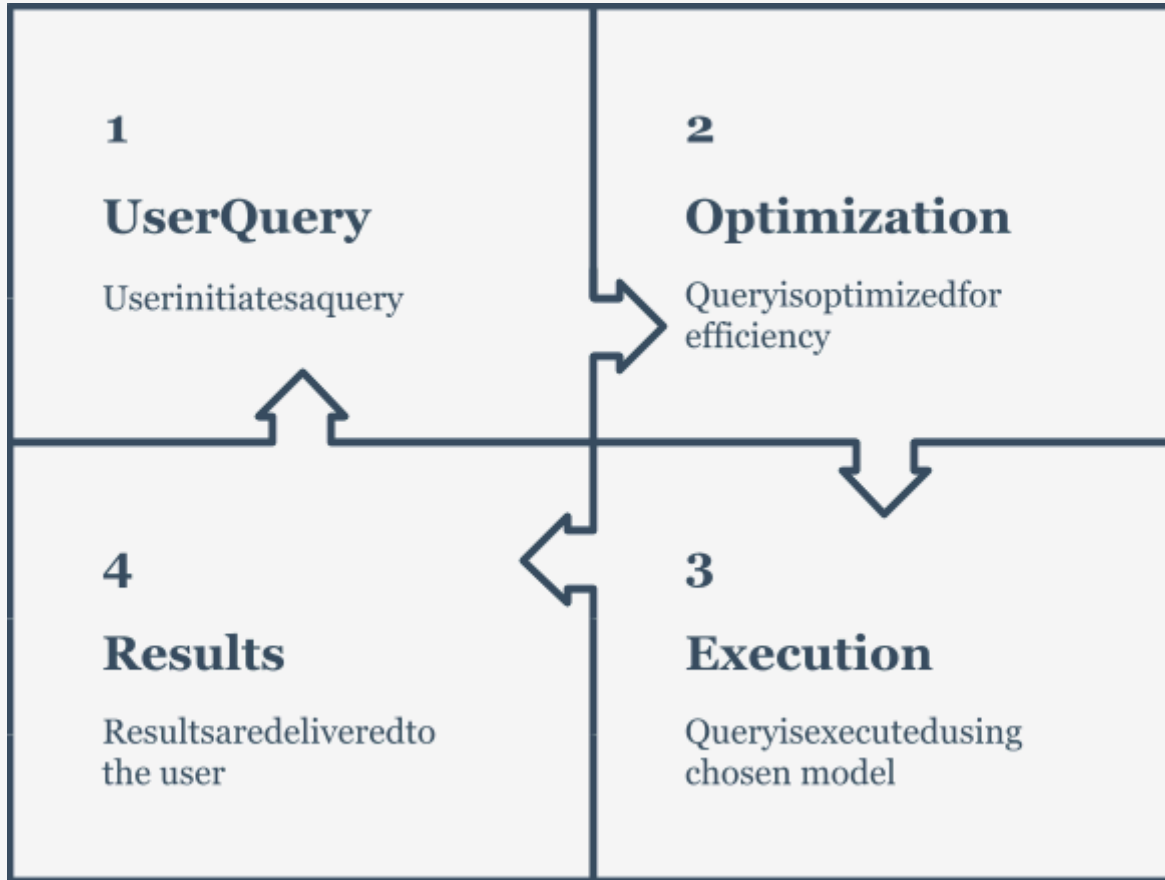
HighData

LowData

StarSchemavs.Snowflake Schema

StarSchema: A central Fact Table connected to Dimension Tables.
Snowflake Schema: Normalized Dimension Tables.



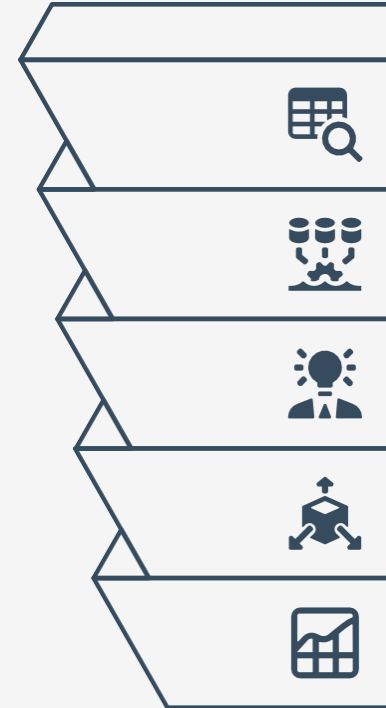


Query Processing & Computation

Computation Models: MOLAP (Multidimensional OLAP), ROLAP (Relational OLAP), and HOLAP (Hybrid OLAP).

Business Intelligence Tools & Workflows

Modern Software and the
KNIMEAnalyticalProcess



Turning Data into Smart Business Decisions



Interactive Dashboards

Real-time monitoring of business health.



KPIs

Tracking Key Performance Indicators.



Data Visualization

Making complex data understandable.



Predictive Analytics

Forecasting future trends.



Smart Business Decisions

The ultimate goal of the ecosystem.